

Week

8

Advanced Git Technique using GitHub:

- What is GitHub
- Setting up GitHub
- Push, Pull and Fetch





**What is
GitHub?**



GitHub

GitHub

A vast platform reshaping how developers collaborate on projects by offering essential tools for version control, issue tracking, and code review.

GitHub

- a platform to host git code repositories
- <http://github.com>
- launched in 2008
- most popular Git host
- allows users to collaborate on projects from anywhere
- GitHub makes git social!
- Free to start



GitHub

1. It is a software

2. It is installed locally on the system

3. It is a command line tool

4. It is a tool to manage different versions of edits, made to files in a git repository

5. It provides functionalities like Version Control System Source Code Management

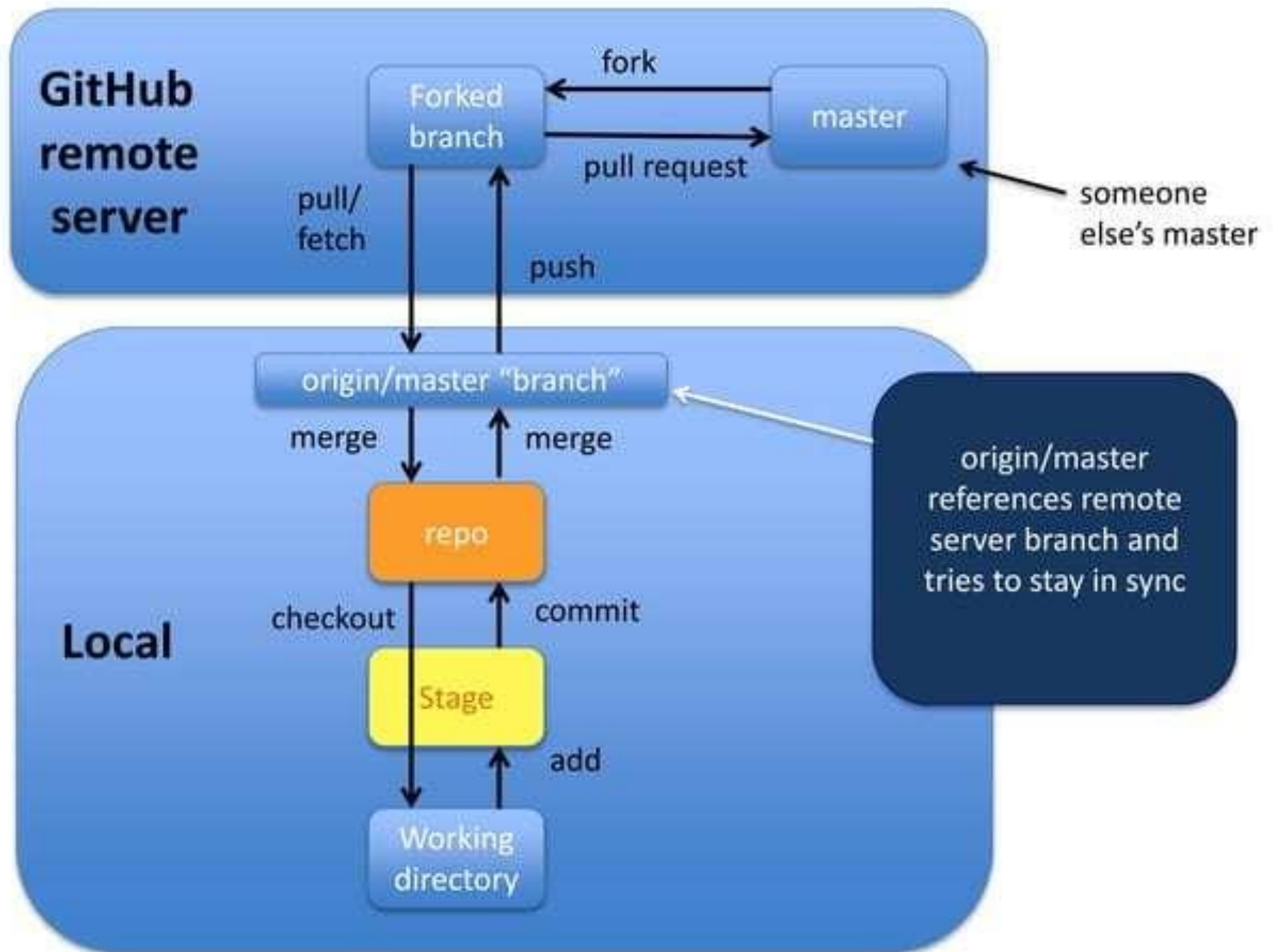
1. It is a service

2. It is hosted on Web

3. It provides a graphical interface

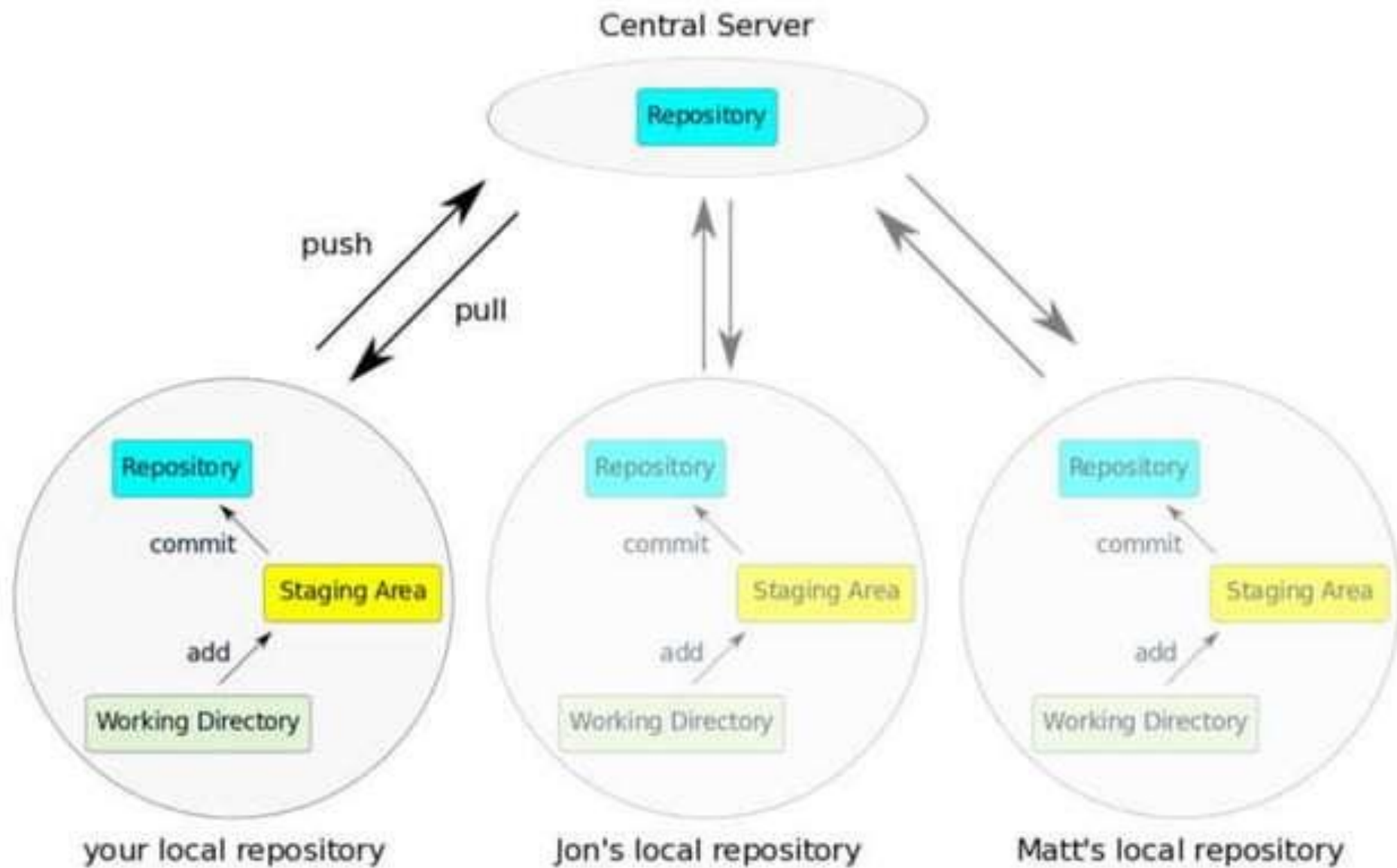
4. It is a space to upload a copy of the **Git** repository

5. It provides functionalities of Git like VCS, Source Code Management as well as adding few of its own features



Important to remember

Sometimes developers *choose* to place repo on GitHub as a centralized place where everyone commits changes, but it **doesn't have to be on GitHub**



Get Started With GitHub





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Let's build from here

The world's leading AI-powered developer platform.

Email address

[Sign up for GitHub](#)

[Start a free enterprise trial >](#)

Welcome to GitHub!

Let's begin the adventure

Enter your email*

✓ christianmichael.mansueto@umak.edu.ph

Create a password*

✓ [12 dots]

Enter a username*

→ CidolfusClive

Continue

CidolfusClive is available.

Welcome to GitHub!

Let's begin the adventure

Verify your account

Use the arrows to rotate the object to face in the direction of the hand. (1 of 1)



Submit



Audio



Restart



Sign in to GitHub

Your account was created
successfully. Please sign in to
continue



Username or email address

Password

[Forgot password?](#)

Sign in

[Sign in with a passkey](#)

New to GitHub? [Create an account](#)



Welcome to GitHub

We are glad you're here.

This will help us guide you to the tools that are best suited for your projects.

How would you describe yourself?

☐ N/A

☐ Student

☒ Teacher

How many team members will be working with you?

☒ Just me

☐ 2-5

☐ 5-10

☐ 10-20

☐ 20-50

☐ 50+

Continue



Tell us a bit about yourself.

We'll get you started on the right track.

What are the top 2 things you want to do with GitHub?

☒ Start a new project

☐ Start or expand my business

☐ Use GitHub Copilot

☐ Connect with other developers

☒ Contribute to Open Source projects

Select up to 2 options

[Continue](#)

/ to search



New repository



Import repository



New codespace



New gist



New organization



New project

Explore repository



facebook / f

An open-source C+
Facebook.



28.2k



sammycage

LunaSVG is a lightweight library for efficient rendering
and manipulation of SVG files.



862

Q Type  to search

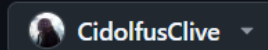


Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Required fields are marked with an asterisk ().*

Owner *



Repository name *

/ DemoDevOps

✔ DemoDevOps is available.

Great repository names are short and memorable. Need inspiration? How about **congenial-train** ?

Description (optional)

This is a demo project for my students.



Public

Anyone on the internet can see this repository. You choose who can commit.



Private

You choose who can see and commit to this repository.

Initialize this repository with:



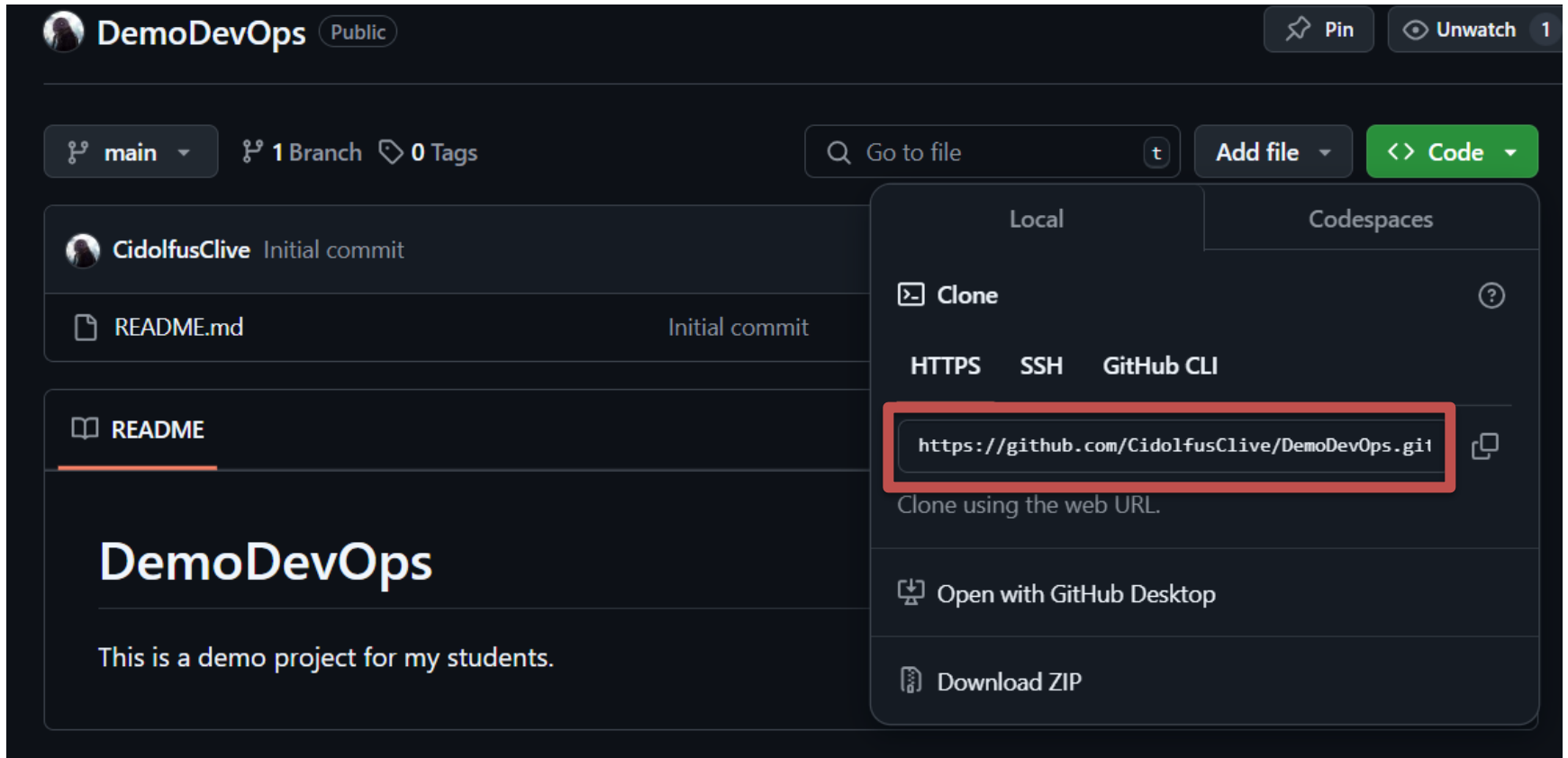
Add a README file

This is where you can write a long description for your project. [Learn more about READMEs.](#)

Add .gitignore

.gitignore template: None

Copying (Cloning) files from remote repo to local machine



The screenshot shows the GitHub interface for a repository named 'DemoDevOps' by user 'CidolfusClive'. The repository is public and has 1 branch and 0 tags. The 'main' branch is selected. The repository contains a 'README.md' file, which is the 'Initial commit'. The 'README' file is open, showing the title 'DemoDevOps' and the description 'This is a demo project for my students.'.

A dropdown menu is open, showing options for cloning the repository. The 'Local' tab is selected, and the 'Clone' button is highlighted. The 'Clone' button has a red border around it. The 'Clone' button is labeled 'Clone' and has a question mark icon. Below the 'Clone' button, there are three options: 'HTTPS', 'SSH', and 'GitHub CLI'. The 'HTTPS' option is selected, and the URL 'https://github.com/CidolfusClive/DemoDevOps.git' is displayed. The URL is highlighted with a red border. Below the URL, it says 'Clone using the web URL.'.

Other options in the dropdown menu include 'Open with GitHub Desktop' and 'Download ZIP'.

Copying (Cloning) files from remote repo to local machine

git clone <URL>

```
chris@cmmansueto MINGW64 ~  
$ git clone https://github.com/CidoIfusClive/DemoDevOps.git  
Cloning into 'DemoDevOps'...  
remote: Enumerating objects: 3, done.  
remote: Counting objects: 100% (3/3), done.  
remote: Compressing objects: 100% (2/2), done.  
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
Receiving objects: 100% (3/3), done.
```

Determine the origin of a cloned Git repository

```
chris@cmmansueto MINGW64 ~/DemoDevOps (main)
$ git ls-remote --get-url origin
https://github.com/CidolfusClive/DemoDevOps.git

chris@cmmansueto MINGW64 ~/DemoDevOps (main)
$ git remote show origin
* remote origin
  Fetch URL: https://github.com/CidolfusClive/DemoDevOps.git
  Push URL: https://github.com/CidolfusClive/DemoDevOps.git
  HEAD branch: main
  Remote branch:
    main tracked
  Local branch configured for 'git pull':
    main merges with remote main
  Local ref configured for 'git push':
    main pushes to main (up to date)

chris@cmmansueto MINGW64 ~/DemoDevOps (main)
$ git remote -v
origin https://github.com/CidolfusClive/DemoDevOps.git (fetch)
origin https://github.com/CidolfusClive/DemoDevOps.git (push)
```

Adding some files...

```
chris@cmmansueto MINGW64 ~/DemoDevOps (main)
```

```
$ git status
```

```
On branch main
```

```
Your branch is up to date with 'origin/main'.
```

```
Untracked files:
```

```
(use "git add <file>..." to include in what will be committed)
```

```
Car.java
```

```
Demo.java
```

```
Person.java
```

```
Student.java
```

```
TestStudentClass.java
```

```
nothing added to commit but untracked files present (use "git add" to track)
```

```
chris@cmmansueto MINGW64 ~/DemoDevOps (main)
```

```
$ git add -A
```

```
chris@cmmansueto MINGW64 ~/DemoDevOps (main)
```

```
$ git commit -m "new added java files"
```

```
[main a6bfbc0] new added java files
```

```
5 files changed, 195 insertions(+)
```

```
create mode 100644 Car.java
```

```
create mode 100644 Demo.java
```

```
create mode 100644 Person.java
```

```
create mode 100644 Student.java
```

```
create mode 100644 TestStudentClass.java
```

Pushing to remote repo

```
chris@cmmansueto MINGW64 ~/DemoDevOps (main)
$ git push
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.83 KiB | 1.83 MiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/CidoIfusClive/DemoDevOps.git
    ec7b2eb..a6bfbc0  main -> main
```

Pushing to remote repo

```
chris@cmmansueto MINGW64 ~/DemoDevOps (main)
$ git log
commit a6bfb0c07d93c223c6aa864d46b5e3a865f12 (HEAD -> main, origin/main, origin/HEAD)
Author: cmmansueto <christianmichael.mansueto@umak.edu.ph>
Date:   Wed Oct 2 15:16:33 2024 +0800

    new added java files

commit ec7b2eb29ca80d78e99ebe1ec024e1fa6a60ebf0
Author: CidolfusClive <christianmichael.mansueto@umak.edu.ph>
Date:   Wed Oct 2 14:54:59 2024 +0800

    Initial commit
```

Pushing to remote repo

 **DemoDevOps** PublicPin Unwatch 1

main 1 Branch 0 Tags Add file Code

 **CidolfusClive** new added java files a6bfb0 · 5 minutes ago 2 Commits

 Car.java	new added java files	5 minutes ago
 Demo.java	new added java files	5 minutes ago
 Person.java	new added java files	5 minutes ago
 README.md	Initial commit	27 minutes ago
 Student.java	new added java files	5 minutes ago
 TestStudentClass.java	new added java files	5 minutes ago

README 

DemoDevOps

This is a demo project for my students.

Creating a branch and adding a new file

```
chris@cmmansueto MINGW64 ~/DemoDevOps (main)
$ git branch
  CM_features
* main

chris@cmmansueto MINGW64 ~/DemoDevOps (main)
$ git checkout CM_features
Switched to branch 'CM_features'
```

```
chris@cmmansueto MINGW64 ~/DemoDevOps (CM_features)
$ git add Test.txt

chris@cmmansueto MINGW64 ~/DemoDevOps (CM_features)
$ git commit -m "added new text file"
[CM_features e91e2f9] added new text file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 Test.txt
```

Creating a branch from local to remote repo

- The remote repository doesn't have a **CM_features** branch. To create a new branch and push changes, we will use “**CM_features:CM_features**”.

```
chris@cmmansueto MINGW64 ~/DemoDevOps (CM_features)
$ git push origin CM_features:CM_features
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 277 bytes | 277.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'CM_features' on GitHub by visiting:
remote:   https://github.com/CidolfusClive/DemoDevOps/pull/new/CM_features
remote:
To https://github.com/CidolfusClive/DemoDevOps.git
 * [new branch]      CM_features -> CM_features
```

Creating a branch from local to remote repo

 **DemoDevOps** Public Pin Unwatch

 **CM_features** had recent pushes 12 seconds ago Compare & pull request

 **main**  **2 Branches**  **0 Tags** t Add file <> Code

 **CidolfusClive** new added java files a6bfbc0 · 28 minutes ago 🕒 2 Commits

 Car.java	new added java files	28 minutes ago
 Demo.java	new added java files	28 minutes ago
 Person.java	new added java files	28 minutes ago
 README.md	Initial commit	50 minutes ago
 Student.java	new added java files	28 minutes ago
 TestStudentClass.java	new added java files	28 minutes ago

Pull Request

- This functionality is common for organizations. For example, a software developer has worked on a new feature and wants to merge changes to the main remote branch. We will now create pull requests using GitHub GUI by clicking on the pull request button. You can type a detailed explanation of what features were added and click on the pull request button.

Pull Request

Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff compa](#)



base: main



compare: CM_features

✓ Able to merge. These branches can be automatically merged.



Add a title

added new text file

Add a description

Write

Preview

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↩

📌

added new text file

Merge pull request on GitHub

added new text file #1

 Open CidolfusClive wants to merge 1 commit into `main` from `CM_features` 

 Conversation 0  Commits 1  Checks 0  Files changed 1

 CidolfusClive commented now Owner ...

added new text file



  added new text file e91e2f9

  **Require approval from specific reviewers before merging** Add rule ×
[Rulesets](#) ensure specific people approve pull requests before they're merged.

 **Continuous integration has not been set up**
[GitHub Actions](#) and [several other apps](#) can be used to automatically catch bugs and enforce style.

 **This branch has no conflicts with the base branch**
Merging can be performed automatically.

Merge pull request  You can also [open this in GitHub Desktop](#) or view [command line instructions](#).

Merge pull request on GitHub

added new text file #1

 Merged CidolfusClive merged 1 commit into `main` from `CM_features`  now

 Conversation 0  Commits 1  Checks 0  Files changed 1

 CidolfusClive commented 4 minutes ago Owner ...

added new text file



  added new text file e91e2f9

  CidolfusClive merged commit `441c08b` into `main` now Revert

 Pull request successfully merged and closed Delete branch

You're all set—the `CM_features` branch can be safely deleted.

Updating a local repository with changes from a GitHub repository

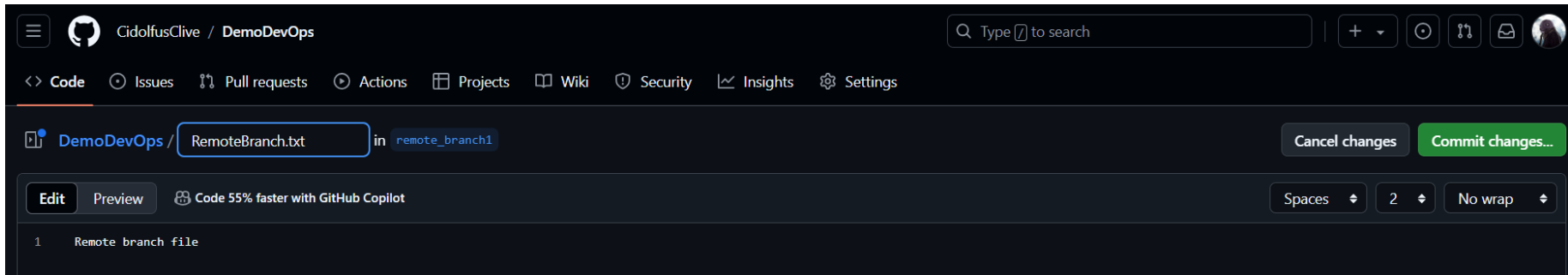
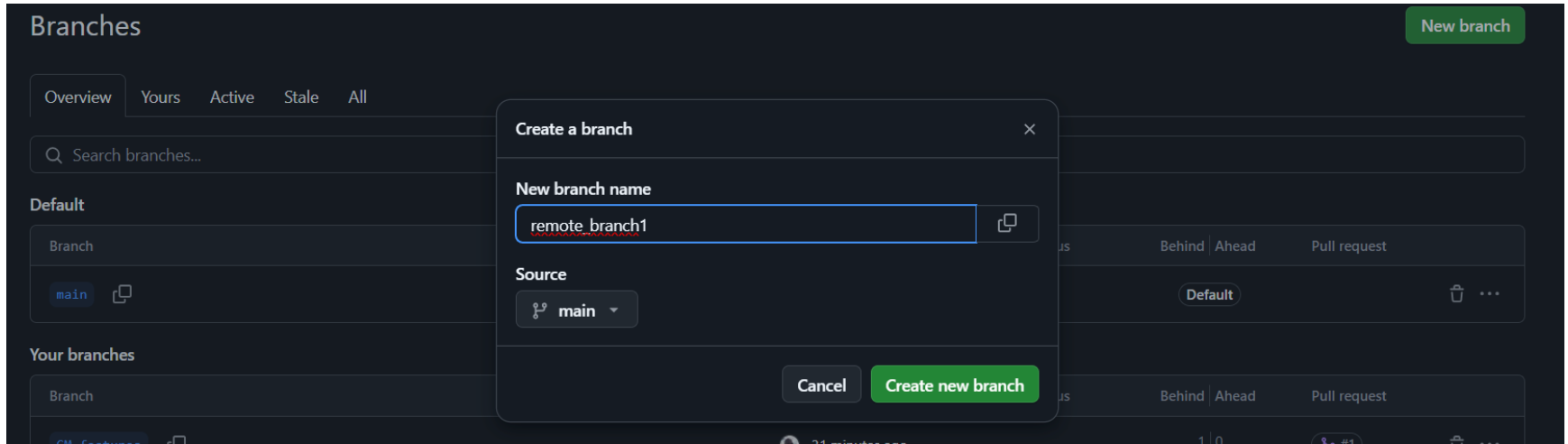
```
chris@cmmansueto MINGW64 ~/DemoDevOps (CM_features)
$ git checkout main
Switched to branch 'main'
Your branch is behind 'origin/main' by 2 commits, and can be fast-forwarded.
(use "git pull" to update your local branch)
```

```
chris@cmmansueto MINGW64 ~/DemoDevOps (main)
$ ls
Car.java  Demo.java  Person.java  README.md  Student.java  TestStudentClass.java

chris@cmmansueto MINGW64 ~/DemoDevOps (main)
$ git pull origin main
From https://github.com/CidolfusClive/DemoDevOps
 * branch          main          -> FETCH_HEAD
Updating a6bfbc0..441c08b
Fast-forward
 Test.txt | 0
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Test.txt

chris@cmmansueto MINGW64 ~/DemoDevOps (main)
$ ls
Car.java  Demo.java  Person.java  README.md  Student.java  Test.txt  TestStudentClass.java
```

Creating a new branch from GitHub



```
chris@cmmansueto MINGW64 ~/DemoDevOps (main)
$ git fetch origin
```

```
chris@cmmansueto MINGW64 ~/DemoDevOps (main)
$ git branch -r
  origin/CM_features
  origin/HEAD -> origin/main
  origin/main
  origin/remote_branch1
```

```
chris@cmmansueto MINGW64 ~/DemoDevOps (main)
$ git branch
  CM_features
* main
```

Synchronize The Local Repository

```
chris@cmmansueto MINGW64 ~/DemoDevOps (main)
```

```
$ git merge origin/main
```

```
Updating 441c08b..2c49e8e
```

```
Fast-forward
```

```
RemoteBranch.txt | 1 +
```

```
1 file changed, 1 insertion(+)
```

```
create mode 100644 RemoteBranch.txt
```

```
chris@cmmansueto MINGW64 ~/DemoDevOps (main)
```

```
$ ls
```

```
Car.java Demo.java Person.java README.md RemoteBranch.txt Student.java Test.txt TestStudentClass.java
```

Pushing to a remote repo

`git push` local_branch_alias branch_name

```
[dolanmi]$ git push origin master
Counting objects: 3, done.
Delta compression using up to 8 threads.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 280 bytes | 0 bytes/s, done.
Total 3 (delta 2), reused 0 (delta 0)
To https://github.com/mchldln/open-sourcerer.git
   ecd0d3b..212432e  master -> master
```

git fetch Vs. git pull

The table below outlines how the `git fetch` and `git pull` commands compare.

`git fetch`

Downloads new information from a remote repository without merging into the current branch.

Updates the repository data in the `.git` directory.

Allows reviewing commits and changes from other developers before committing the code.

No possibility of merge conflicts when running the command.

`git pull`

Downloads all the changes from a remote repository and merges them into the current branch.

Updates the local repository directly.

Allows bringing and updating changes to the local repository immediately.

Possible merge conflicts which need an immediate resolution.

How do I link my local repo to a remote repo?

```
git remote add <alias> <URL>
```

Note: This just establishes a connection...no files are copied/moved

Note: Yes! You may have more than one remote linked to your local directory!

Which remotes am I linked to?

```
git remote
```

Fetching from a remote repo

```
git fetch remote_repo_name
```

Fetch in no way changes a your working dir or any commits that you've made.

- Fetch before you work
- Fetch before you push
- Fetch often

git merge must be done to merge fetched changes into local branch

Good resources

- Git from Git: <https://git-scm.com/book/en/v2>
- A guided tour that walks through the fundamentals of Git:
<https://githowto.com>
- Linus Torvalds on Git:
<https://www.youtube.com/watch?v=idLyobOhtO4>
- Git tutorial from Atlassian:
<https://www.atlassian.com/git/tutorials/>
- A number of easy-to-understand guides by the GitHub folks
<https://guides.github.com>

git commit -a

- Allows one to add to staging index and commit *at the same time*
- Grabs everything in working directory
- Files not tracked or being deleted are not included

`git diff g5iU0oPe7x`

When using checksum of older commit, will show you all changes compared to those in your working directory

Renaming and deleting branches

```
git branch -m/--move old_name new_name
```

```
git branch -d branch_name
```

Note: Must not be in branch_name

Note: Must not have commits in branch_name unmerged in branch from which you are deleting

```
git branch -D branch_name
```

Note: If you are **really** sure that you want to delete branch with commits

Tagging

- Git has the ability to tag specific points in history as being important, such as releases versions (v.1.0, 2.0, ...)

git tag

```
$ git tag  
v0.1  
v1.3
```

Tagging

Two types of tags:

lightweight – a pointer to a specific commit –
basically a SHA stored in a file

```
git tag tag_name
```

annotated – a full object stored in the Git database –
SHA, tagger name, email, date, message
and can be signed and verified with GNU
Privacy Guard (GPG)

```
git tag -a tag_name -m "message"
```

How do I see tags?

git show tag_name

```
$ git show v1.4-lw
commit ca82a6dff817ec66f44342007202690a93763949
Author: Scott Chacon <schacon@gee-mail.com>
Date: Mon Mar 17 21:52:11 2008 -0700
```

changed the version number

Lightweight tag

```
$ git show v1.4
tag v1.4
Tagger: Ben Straub <ben@straub.cc>
Date: Sat May 3 20:19:12 2014 -0700
```

my version 1.4

```
commit ca82a6dff817ec66f44342007202690a93763949
Author: Scott Chacon <schacon@gee-mail.com>
Date: Mon Mar 17 21:52:11 2008 -0700
```

changed the version number

Annotated tag